CVPR #*****

Confidence for Which Prediction? Supervision, Readout, and Coverage in Visual Grounding Supplementary Material

Anonymous CVPR submission

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A. Coverage Interpretation of the Existing Results

The revision elevates two existing diagnostics: inference-only versus matched-training readout changes, and L1 versus cross-difficulty state comparisons. No model, score record, evaluation surface or bootstrap draw is added. The following consequence is analytic rather than a new estimate of a conditional risk curve.

For the MM-GDINO FineCops global emission-minus-existence contrast, the L1 pairwise effects are +7.085 [+6.023, +8.236] for C/W and +0.224 [+0.024, +0.431] for C/N. The positive class in this conditional population has 5,506 correct and 591 wrong outputs, while its negative class contains all 9,029 existing no-target requests. Native correctness is fixed across head seeds, so $a_1 = 5{,}506/6{,}097$ is common to the three comparisons.

With bars denoting the fixed equal-seed mean, for $0 \leq \pi < 1$,

$$\overline{\Delta R_1}(\pi) = -(1-\pi)a_1\big[(1-\pi)(1-a_1)\overline{\Delta U^{(1)}_{CW}} + \pi\overline{\Delta U^{(1)}_{CN}}\big] < 0. \quad (1)$$

Both displayed point differences are positive, so every non-degenerate mixture has a lower mean AUGRC for emission; at $\pi = 1$ the difference is zero. There is no interior crossover of this mean curve. Linearity in the pairwise effects justifies using their equal-seed means, rather than pooling scores from different heads.

The intervals attached to the two pairwise effects are the existing image-bootstrap intervals. They are not a newly computed L1 risk interval or a simultaneous band over π. The sign statement is a post-hoc analytic implication of fixed means, not additional training, a new coverage intervention, or evidence that the effect is necessarily caused by lack of L2/L3 supervision. L1 describes difficulty support; it does not imply that these validation requests were training examples.

The full C/N effect is the sum of a weighted within-level contribution, +0.169 [+0.019, +0.326], and a cross-level contribution, -3.692 [-4.019, -3.384]. The conditional L1 C/N AUROC change +0.224 [+0.024, +0.431] is not itself the weighted contribution. Every no-target expression has an L1 positive parent, so cross-level pairs here involve L2/L3 positives and L1-parent negatives; the negative edit level is not substituted for positive-expression difficulty.

B. What Is Controlled

Within each frozen localizer, the experiment crosses two labels (target existence and correct Native emission) with two query readouts (global max and Native-selected query). The output box, query features, native scores, head parameterization, initialization and sample schedule are unchanged within each comparison. The readout intervention jointly changes the scored query, competition among queries, and the location of training gradients. It is not a single-factor manipulation of geometric correspondence.

The MM-GDINO global heads are the original fixed-target endpoints. The six selected MM-GDINO heads and twelve MDETR heads are new, all trained with seeds 17/42/73 to the same fixed endpoint. No original checkpoint or record is rewritten, and no result selects the subsequent model matrix. The three seeds refer to confidence heads, not independent localizer fine-tuning runs.

C. Localizers, Native Selection and Data Lineage

C.1. MM-GDINO-T

The frozen model is the FineCops positive-source epoch-five artifact initialized directly from the released MM-GDINO-T pretrain, not from a RefCOCO-finetuned checkpoint. Its original fine-tuning used one GPU, physical batch four and accumulation eight for effective batch 32. The directory's historical batch label is not a statement that the trunk used physical batch 32.

Table 1. The MM-GDINO FineCops global target effect changes across difficulty comparisons. All numbers are emission-minus-existence AUROC points with existing paired image intervals. L1 uses 6,097 positive requests and the same 9,029 no-target requests. The final two rows are weighted contributions to the full C/N effect, not conditional AUROCs.

Comparison	ΔU_{CW}	ΔU_{CN}
Full positive population	+11.654 [+10.721, +12.641]	-3.523 [-3.884, -3.166]
L1 positive population	+7.085 [+6.023, +8.236]	+0.224 [+0.024, +0.431]
Within-level contribution	–	+0.169 [+0.019, +0.326]
Cross-level contribution	–	-3.692 [-4.019, -3.384]

Table 2. Two ways to read the Native-selected query on MDETR FineCops. Both changes reference the G-trained/G-deployed emission head. Matched S is trained from the same original initialization, not continued from G. Risk is mixed AUGRC times 100; endpoint cells show seed mean (sample SD), differences show paired image intervals for the fixed three-head mean.

Training	Inference	Risk (SD)	Change from G/G
Global	Global	28.84 (0.13)	–
Global	Selected	30.45 (2.83)	+1.608 [+1.513, +1.698]
Selected	Selected	28.65 (0.02)	-0.189 [-0.257, -0.122]

The 83,341 positive-source rows were optimized with ordinary FP32 AdamW: LR 2×10^{-4}, backbone multiplier 0.1, weight decay 10^{-4}, gradient clipping 0.1 and epoch-three LR multiplier 0.1. The seed was 20260902 with deterministic training disabled and no frozen detector modules. No-flip multiscale resize used short sides 480–800 and max 1333, or an intermediate 400/500/600 resize, an absolute-range crop allowing empty crops, and the final multiscale resize. Positive-source does not mean the absence of background supervision: unmatched queries and permissible empty crops retain negative signals.

We reuse the fixed epoch-five weights checked through frozen runtime/candidate parity, rather than training another localizer. The current study does not reopen FineCops Test.

Frozen extraction uses the full expression, single-image inference, 800/max1333 resize, and all 900 final-layer 256-d queries. Native scores are maximum sigmoid token scores; the maximum Native-score query wins, with the original lowest-index tie rule.

C.2. MDETR-R101

The second localizer is the official RefCOCO R101 EMA release of MDETR [1]. The pinned implementation and weights are bound by the study manifests. The required checkpoint key is `model_ema`; missing EMA does not trigger fallback to raw weights. This is a RefCOCO-adapted second representation, not a FineCops-matched architecture-only comparison.

The complete expression and official preprocessing produce 100 256-d query features. Native score is $1 - \mathrm{softmax}(\mathrm{pred\_logits})_{\mathrm{no\ object}}$. Selection preserves the official descending score–pixel-xyxy tuple ordering; completely duplicated tuples retain the lowest original query index. Candidates are neither padded to 900 nor postprocessed with a new NMS or confidence threshold. Correctness is recomputed from MDETR's own selected box using IoU at least 0.5. An upstream evaluator's GIoU aggregate is not substituted for this study's IoU metric.

C.3. Populations and image overlap

FineCops train contains 83,341 unique positive-source requests and 80,451 paired negative-text requests. Every paired positive parent is L1: the 80,451 confidence pairs reference 43,979 unique L1 positives. No L2/L3 positive enters either target's head loss, despite being present in the frozen cache. The full 83,341-positive pool used for SIRC normalization must not be described as full positive head-supervision coverage. MM-GDINO's earlier positive-source trunk adaptation did use the full positive population. Validation includes 9,426 positives and 9,029 text negatives on 3,567 images. Those validation negatives reference 4,946 unique positive parents, also all L1; the validation positive population nevertheless spans L1–L3. No new head is evaluated on FineCops Test or negative-image perturbations.

Thus 10,036 other L1 positives and all 29,326 L2/L3 training positives receive no confidence-head loss. Generating a permutation over all positives does not change

Table 3. Positive-request coverage by original expression difficulty. Pair counts include repeated parents; unique-positive counts do not. SIRC normalization sees every training positive but supplies no head-loss updates.

Population	L1	L2	L3
Train unique positives	54,015	25,282	4,044
Train unique head-loss parents	43,979	0	0
Train confidence pairs	80,451	0	0
Train positives outside head loss	10,036	25,282	4,044
Val positives	6,097	2,884	445
Val unique negative parents	4,946	0	0
Val negative/parent pairs	9,029	0	0

this: those are rank events, and the head trainer skips every rank event. Only the confidence-pair events update the heads. The complete unique-positive pool enters SIRC score-statistic normalization, which is separate from supervised optimization.

This boundary is shared by all target/readout arms and both localizers. It does not invalidate the label intervention on its fixed sampled source, but it prevents an inference that the heads were directly taught correctness on the full training difficulty range or that evaluation complexity is matched. The observed effects can include generalization outside positive head-loss coverage; these data do not identify coverage as the sole or causal explanation of any performance difference.

gRefCOCO TestA contains 5,917 single-target positives and 4,448 no-targets; TestB contains 5,646 and 4,673. Their pooled surface has 20,684 requests on 1,500 images. All 14,579 multi-target expressions are excluded, so this is not full zero-to-many-target GREC. The source-disjoint sensitivity excludes FineCops training and validation source images through VG–COCO identities or exact bytes: 202 train and 21 validation images are removed, leaving 1,277 images, 9,848 positives and 7,716 no-targets. Global pretraining exposure and near duplicates remain unenumerated. Full and source-disjoint overlap and are not independent benchmarks. MDETR is RefCOCO-adapted, and gRefCOCO is built on COCO imagery. The current FineCops-source exclusion is not a separate image-disjointness audit of MDETR's complete pretraining/adaptation lineage.

The mechanism protocol was locked before new readout training/evaluation, but motivated by earlier results on previously investigated benchmarks. This is an exploratory mechanism study with a prospectively fixed recipe, not a virgin held-out confirmation or an image-disjoint zero-shot claim.

D. Head Architecture, Optimization and Resume

For every valid query, the head input is the detached 256-d feature concatenated with the detached Native score. LayerNorm(257), Linear(257,128), GELU, Linear(128,128), GELU and Linear(128,1), including biases, give 50,179 parameters in eight tensors. The final weight and bias start at zero. The original factory creates the full historical shell before copying initialization to each arm, retaining its random-number consumption. No selected head is initialized from a trained global endpoint.

The fixed training score is c_G or c_S in both the loss and matched inference. If p_i, n_i denote paired source logits,

$$
\begin{aligned}
L = {} & \tfrac{1}{2}\overline{\mathrm{BCE}(p_i, y_i)} + \tfrac{1}{2}\overline{\mathrm{BCE}(n_i, 0)} \\
& + \tfrac{10^{-3}}{2}\big(\overline{p_i^2} + \overline{n_i^2}\big). \qquad (2)
\end{aligned}
$$

Existence sets $y_i = 1$ on every sampled positive parent; emission uses fixed Native correctness. Neither target changes source weights, sample frequency or target-dependent class balance. A target logit is an ordering score, not an asserted calibrated conditional probability.

AdamW uses LR 10^{-4}, zero weight decay, gradient clip 0.1 and deterministic FP32, with 32 pairs per update. The original event generator draws the rank permutation first, then the confidence permutation; rank events are generated and skipped rather than removed from its RNG history. Each epoch has 2,514 full batches and one three-pair batch, giving 2,515 updates per epoch and 12,575 after five epochs. Both readouts compute full dense logits before max or gather. The skipped rank events provide neither rank nor confidence supervision to the unpaired L1 or L2/L3 positives.

Independent optimizer states, model parameters, RNG state and population/ code bindings are serialized at every epoch. Tests cover equality of the legacy global trajectory and serialized interrupted/resumed trajectories. Frozen rank tensors stay bitwise unchanged; selected loss has no gradient on unselected dense logits, and neither head has an autograd connection to the frozen query source or Native scores. No AMP scaler or skipped optimizer update is used.

E. Inference Groups and Geometry Diagnostics

Cached features are FP16; Native scores and boxes are FP32. Heads cast features to FP32, keeping this precision boundary identical to training. The val evaluator preserves contiguous batches of 32 and its final batch of 23. gRef preserves the original four-worker source-index-modulo-four assignment, 32-request head batches and final batch of 19 per worker. This grouping matters for exact score reproduction even when output differences from another grouping would be numerically tiny.

Every gRef localizer is traversed once and cached for all heads. The complete 18-head endpoint gate precedes any new transfer scoring. Old MM-GDINO global logits and Native query/box/hash records must match bitwise. No-target has null correctness and GT IoU; it never acquires a fabricated box label. Confidence never changes Native geometry.

For each trained head we save its global maximum and selected-query logit, the confidence winner and Native indices, their box IoU, and both GT IoUs where a true target exists. Different indices can have similar boxes. Throughout the winner tables, *winner* means the *global* argmax of that head's dense logits. It supplies deployed confidence for G; for S it is only a counterfactual diagnostic. Matched S deployment always reads the Native-selected query, so its actual readout-index disagreement is identically zero. Conversely, a Native-wrong request can have a correct confidence-winning query, supporting existence evidence without certifying its emitted box. The tables report these events descriptively, not as uniquely attributable causal fractions.

F. Three-State Risk Identities

Let C, W, N denote correct emission, wrong positive localization and no-target, with masses $p_C = (1-\pi)a$, $p_W = (1-\pi)(1-a)$ and $p_N = \pi$. With half credit for score ties,

$$\mathrm{AUC}_E = aU_{CN} + (1-a)U_{WN}. \tag{3}$$

For generalized-risk integration starting at $(0, 0)$,

$$\begin{aligned}\mathrm{AUGRC} = {} & p_C p_W(1-U_{CW}) + p_C p_N(1-U_{CN}) \\ & + \tfrac{1}{2}(p_W + p_N)^2. \end{aligned} \tag{4}$$

The last term is fixed within a localizer/population. Taking differences therefore yields the main-paper identity without a W/N term. The identity holds with tied scores and fractional request weights. All-correct and all-failure AUGRC limits are 0 and $1/2$; all-tied AUGRC is half the failure rate. We report AUGRC and AURC multiplied by 100, despite their different ranges.

The fixed prior grid is $\{0, .1, .25, .5, .75, .9, 1\}$. Every draw separately normalizes positive and no-target request masses to $1-\pi$ and π. Observed-mixture analysis instead retains the image draw's empirical class proportions. Mixing endpoint integrals linearly would not reconstruct the reweighted ranking curve.

For AURC, entire equal-score groups are admitted together. Trapezoids integrate selective risk between successive achieved coverages, extending the first nonempty group's risk to zero coverage. AUGRC instead starts at zero generalized risk and zero coverage. At requested coverages 10/25/50/75/90/100%, the boundary tie group is retained and the achieved coverage is reported. Wrong-box and no-target contributions sum exactly under both integrals. These are curve diagnostics, not selected deployment thresholds.

For score differences ΔU_{CW} and ΔU_{CN}, an interior AUGRC crossing solves $(1-\pi)(1-a)\Delta U_{CW} + \pi \Delta U_{CN} = 0$. We solve for the equal-seed mean risk difference, not the average of per-seed roots. If any replicate has no interior crossing, an unconditional interior-root interval is not reported. Interior-only intervals are explicitly conditional and accompanied by counts of every other root status.

G. Conditional Comparisons and Uncertainty

Same-image pairwise AUROCs include all eligible state pairs within participating images. Under an image draw of multiplicity m_i, those within-copy pairs contribute m_i, not m_i^2. Their comparable unconditional analysis uses precisely the same participating requests. This permits a paired attenuation contrast without silently replacing the evaluation population.

FineCops negative records point to their actual positive parents. Parent-pair wins are reported separately when the parent is Native-correct or Native-wrong. For difficulty conditioning, negative records inherit the *original positive expression's* level; their edit level is kept separate and never treated as positive-expression difficulty. All original negative parents are L1 in both train and validation. Consequently, same-level C/N and W/N conditioning has only an L1 surface; no L2/L3 negatives are invented. Cross-level comparisons with L2/L3 positives also cross the positive head-supervision boundary. The comparable-unconditional difficulty score includes all requests with an original-positive level, whereas the same-level score includes only available within-level state pairs. Their contrast changes pair composition; unlike the same-image analysis, it must not be described as holding the full target distribution matched. Per-level high/low state counts explicitly expose the zero cells. gRef has same-image analysis but no fabricated edited-expression parent.

AUROC can be decomposed into within-level and across-level pair contributions. Their sum is exact, but a contribution share is not causal explained variance. A conditional interval spanning zero does not establish that a whole effect disappears; attenuation needs both a meaningful magnitude and sufficient precision. Undefined comparisons remain undefined rather than becoming zero.

The 5,000 paired image-cluster bootstrap uses PCG64 seed 20260911. The same draw covers every localizer, arm, combination and head seed on a surface; gRef is stratified by TestA/B. Metrics are recomputed per seed and equally averaged. Seed sample SD is reported separately, with no $n = 3$ t-test. An undefined draw invalidates the affected interval;

draws are never silently deleted. Diagnostic FPR95 recomputes the positive q05 in each replicate, accepting equality. No deployment threshold is fitted.

H. Fixed Combination Controls

Product uses $\log \max(s_0, 10^{-6}) + \log \sigma(z_E)$. SIRC-style [2] uses the prespecified rule in the main paper with mean and population SD estimated on all 83,341 unique training positives. Pair-expanded duplicates, validation requests and gRef requests are rejected by its fitting interface. Each localizer/seed has its own bound score-statistics artifact; the same statistics are reused unchanged across evaluation surfaces. If the SD is at most 10^{-12}, the rule returns Native and records the degeneration.

The combinations are evaluated against Native, global existence and global emission, retaining any cost relative to another reference. These controls test two particular score constructions, not optimal fusion or impossibility of combining existence and localization information. Neither alters localization or claims calibrated probability.

I. Complete Numeric Results

The following tables print the prespecified endpoint, seed and core mechanism summaries. The full machine-readable analyses additionally retain every fixed-prior, parent-pair and frozen-weight alternate-readout contrast. Printed conditioning tables focus on the absolute emission readout change and the global target contrast, showing comparable unconditional and conditional estimates as well as their paired difference. Count tables expose each comparison's population. The frozen-weight cross-readout table reports both matched/alternate risks and their paired difference; these off-diagonal diagnostics never replace the matched four-cell matrix. The compact FineCops 50/90%-coverage table reports achieved request coverage and both accepted-error components. Its error probabilities are conditional on acceptance, not counts or fractions of the full request population; they must not be confused with generalized risk.

J. Reusable Evaluation Deliverable

The record interface contains sample/image/stratum identities, localizer-specific fixed-box correctness, score slots and query-winner diagnostics. Source bindings keep checkpoint, cache, initialization, sample order and pairing explicit. Evaluation and statistics remain separate: producing a complete score panel does not choose a model.

Synthetic tests cover the two-target/two-readout gradients, exact interaction algebra, same-image multiplicities, state-risk identities, absent crossover roots, source-disjoint subsets, score ties, replicate q05 recalculation and combination-statistics provenance. The tool can evaluate other localizers and scores from saved records. Its role is a reusable implementation of the study, not an additional general novelty claim.

Table 4. Mixed AUGRC by seed, $\times 100$. Native repeats one frozen prediction route, not three localizer training seeds. Part 1.

Localizer	Population	Score	Seed 17	Seed 42	Seed 73
MM-GDINO-T	FineCops val	Native	28.11	28.11	28.11
MM-GDINO-T	FineCops val	G/E	26.17	26.31	26.21
MM-GDINO-T	FineCops val	G/Y	26.36	26.42	26.35
MM-GDINO-T	FineCops val	S/E	26.12	26.27	26.20
MM-GDINO-T	FineCops val	S/Y	26.32	26.40	26.35
MM-GDINO-T	FineCops val	Product	26.62	26.64	26.59
MM-GDINO-T	FineCops val	SIRC-style	27.37	27.36	27.34
MDETR-R101	FineCops val	Native	30.16	30.16	30.16
MDETR-R101	FineCops val	G/E	29.65	30.00	29.47
MDETR-R101	FineCops val	G/Y	28.98	28.74	28.79
MDETR-R101	FineCops val	S/E	29.54	29.78	29.23
MDETR-R101	FineCops val	S/Y	28.64	28.67	28.65
MDETR-R101	FineCops val	Product	29.25	29.53	29.20
MDETR-R101	FineCops val	SIRC-style	30.01	30.01	30.01
MM-GDINO-T	gRef source-disjoint	Native	30.05	30.05	30.05
MM-GDINO-T	gRef source-disjoint	G/E	28.61	28.76	28.66
MM-GDINO-T	gRef source-disjoint	G/Y	28.61	28.65	28.57
MM-GDINO-T	gRef source-disjoint	S/E	28.53	28.75	28.67
MM-GDINO-T	gRef source-disjoint	S/Y	28.58	28.67	28.65
MM-GDINO-T	gRef source-disjoint	Product	28.49	28.52	28.45
MM-GDINO-T	gRef source-disjoint	SIRC-style	28.72	28.72	28.70
MDETR-R101	gRef source-disjoint	Native	21.13	21.13	21.13
MDETR-R101	gRef source-disjoint	G/E	20.57	20.23	19.52
MDETR-R101	gRef source-disjoint	G/Y	19.54	20.30	19.91

Table 5. Mixed AUGRC by seed, $\times 100$. Native repeats one frozen prediction route, not three localizer training seeds. Part 2.

Localizer	Population	Score	Seed 17	Seed 42	Seed 73
MDETR-R101	gRef source-disjoint	S/E	20.29	20.51	20.23
MDETR-R101	gRef source-disjoint	S/Y	20.41	20.37	20.28
MDETR-R101	gRef source-disjoint	Product	20.26	19.90	19.36
MDETR-R101	gRef source-disjoint	SIRC-style	20.97	20.92	20.96
MM-GDINO-T	gRef Full	Native	30.24	30.24	30.24
MM-GDINO-T	gRef Full	G/E	28.72	28.88	28.77
MM-GDINO-T	gRef Full	G/Y	28.75	28.80	28.70
MM-GDINO-T	gRef Full	S/E	28.65	28.87	28.78
MM-GDINO-T	gRef Full	S/Y	28.71	28.80	28.77
MM-GDINO-T	gRef Full	Product	28.62	28.65	28.58
MM-GDINO-T	gRef Full	SIRC-style	28.87	28.86	28.85
MDETR-R101	gRef Full	Native	21.27	21.27	21.27
MDETR-R101	gRef Full	G/E	20.71	20.37	19.66
MDETR-R101	gRef Full	G/Y	19.70	20.49	20.07
MDETR-R101	gRef Full	S/E	20.46	20.68	20.42
MDETR-R101	gRef Full	S/Y	20.57	20.55	20.46
MDETR-R101	gRef Full	Product	20.41	20.05	19.50
MDETR-R101	gRef Full	SIRC-style	21.11	21.06	21.10

Table 6. Per-seed mixed-AUGRC effects, $\times 100$: fixed seeds 17/42/73, their arithmetic mean and sample SD ($n-1$ denominator), and the paired image-cluster 95% interval of the fixed three-seed mean effect. The image interval is not a training-seed interval: it can exclude zero even when the listed seed effects have different signs. No $n=3$ t-test is used. Part 1.

Localizer	Population	Effect	Seed 17	Seed 42	Seed 73	Mean (SD)	Image CI
MM-GDINO-T	FineCops val	D_Y	-0.041	-0.021	-0.007	-0.023 (0.017)	[-0.048, +0.003]
MM-GDINO-T	FineCops val	D_E	-0.053	-0.041	-0.008	-0.034 (0.023)	[-0.068, +0.000]
MM-GDINO-T	FineCops val	I	+0.012	+0.020	+0.001	+0.011 (0.009)	[-0.016, +0.038]
MM-GDINO-T	FineCops val	$G:Y-E$	+0.182	+0.116	+0.148	+0.149 (0.033)	[+0.058, +0.235]
MM-GDINO-T	gRef source-disjoint	D_Y	-0.029	+0.017	+0.078	+0.022 (0.054)	[-0.013, +0.057]
MM-GDINO-T	gRef source-disjoint	D_E	-0.078	-0.012	+0.010	-0.027 (0.046)	[-0.060, +0.007]
MM-GDINO-T	gRef source-disjoint	I	+0.049	+0.030	+0.068	+0.049 (0.019)	[+0.013, +0.085]
MM-GDINO-T	gRef source-disjoint	$G:Y-E$	-0.001	-0.109	-0.089	-0.066 (0.057)	[-0.178, +0.043]
MM-GDINO-T	gRef Full	D_Y	-0.042	+0.005	+0.071	+0.011 (0.057)	[-0.021, +0.045]
MM-GDINO-T	gRef Full	D_E	-0.071	-0.010	+0.005	-0.025 (0.041)	[-0.055, +0.006]
MM-GDINO-T	gRef Full	I	+0.029	+0.015	+0.066	+0.037 (0.026)	[+0.003, +0.069]
MM-GDINO-T	gRef Full	$G:Y-E$	+0.031	-0.081	-0.070	-0.040 (0.062)	[-0.143, +0.062]

Table 7. Per-seed mixed-AUGRC effects, $\times 100$: fixed seeds 17/42/73, their arithmetic mean and sample SD ($n-1$ denominator), and the paired image-cluster 95% interval of the fixed three-seed mean effect. The image interval is not a training-seed interval: it can exclude zero even when the listed seed effects have different signs. No $n=3$ t-test is used. Part 2.

Localizer	Population	Effect	Seed 17	Seed 42	Seed 73	Mean (SD)	Image CI
MDETR-R101	FineCops val	D_Y	-0.349	-0.075	-0.143	-0.189 (0.143)	[-0.257, -0.122]
MDETR-R101	FineCops val	D_E	-0.105	-0.226	-0.241	-0.191 (0.075)	[-0.285, -0.094]
MDETR-R101	FineCops val	I	-0.245	+0.151	+0.098	+0.001 (0.215)	[-0.091, +0.088]
MDETR-R101	FineCops val	$G:Y-E$	-0.664	-1.261	-0.672	-0.866 (0.342)	[-0.988, -0.745]
MDETR-R101	gRef source-disjoint	D_Y	+0.872	+0.068	+0.371	+0.437 (0.406)	[+0.363, +0.507]
MDETR-R101	gRef source-disjoint	D_E	-0.283	+0.283	+0.708	+0.236 (0.497)	[+0.137, +0.332]
MDETR-R101	gRef source-disjoint	I	+1.155	-0.215	-0.336	+0.201 (0.829)	[+0.108, +0.292]
MDETR-R101	gRef source-disjoint	$G:Y-E$	-1.032	+0.074	+0.392	-0.189 (0.748)	[-0.291, -0.083]
MDETR-R101	gRef Full	D_Y	+0.865	+0.059	+0.395	+0.440 (0.405)	[+0.373, +0.506]
MDETR-R101	gRef Full	D_E	-0.256	+0.305	+0.753	+0.268 (0.506)	[+0.176, +0.356]
MDETR-R101	gRef Full	I	+1.121	-0.247	-0.358	+0.172 (0.824)	[+0.085, +0.259]
MDETR-R101	gRef Full	$G:Y-E$	-1.008	+0.117	+0.405	-0.162 (0.747)	[-0.265, -0.065]

Table 8. All score endpoints, $\times 100$: seed means with paired image-cluster intervals. P@1 is identical across confidence routes. Part 1.

Localizer	Population	Score	P@1	Existence	C/W	FPR95	AURC	AUGRC
MM-GDINO-T	FineCops val	Native	77.36 [76.18, 78.56]	45.36 [44.43, 46.28]	83.28 [82.22, 84.39]	96.09 [95.47, 96.62]	54.57 [53.75, 55.41]	28.11 [27.76, 28.46]
MM-GDINO-T	FineCops val	G/E	77.36 [76.18, 78.56]	62.62 [61.85, 63.39]	68.89 [67.49, 70.30]	80.11 [78.93, 81.21]	47.39 [46.52, 48.25]	26.23 [25.84, 26.60]
MM-GDINO-T	FineCops val	G/Y	77.36 [76.18, 78.56]	55.87 [54.97, 56.76]	80.55 [79.34, 81.74]	85.73 [84.83, 86.67]	47.69 [46.84, 48.52]	26.38 [26.01, 26.74]
MM-GDINO-T	FineCops val	S/E	77.36 [76.18, 78.56]	62.62 [61.83, 63.38]	69.22 [67.83, 70.61]	80.28 [79.15, 81.44]	47.26 [46.40, 48.12]	26.20 [25.81, 26.57]
MM-GDINO-T	FineCops val	S/Y	77.36 [76.18, 78.56]	56.21 [55.31, 57.09]	79.95 [78.74, 81.15]	87.15 [86.20, 88.01]	47.58 [46.74, 48.41]	26.36 [25.99, 26.71]
MM-GDINO-T	FineCops val	Product	77.36 [76.18, 78.56]	55.02 [54.17, 55.88]	80.94 [79.80, 82.04]	81.91 [80.91, 82.84]	48.90 [48.05, 49.73]	26.62 [26.25, 26.97]
MM-GDINO-T	FineCops val	SIRC-style	77.36 [76.18, 78.56]	50.89 [50.01, 51.76]	82.38 [81.30, 83.50]	82.15 [81.26, 83.05]	53.10 [52.26, 53.95]	27.35 [26.99, 27.71]
MDETR-R101	FineCops val	Native	65.56 [64.35, 66.73]	51.56 [50.76, 52.38]	74.62 [73.50, 75.75]	94.37 [93.64, 95.02]	57.59 [56.70, 58.48]	30.16 [29.77, 30.53]
MDETR-R101	FineCops val	G/E	65.56 [64.35, 66.73]	61.55 [60.84, 62.28]	64.90 [63.72, 66.06]	86.01 [85.12, 86.90]	55.23 [54.31, 56.14]	29.71 [29.30, 30.12]
MDETR-R101	FineCops val	G/Y	65.56 [64.35, 66.73]	56.83 [56.04, 57.64]	78.79 [77.79, 79.73]	90.40 [89.64, 91.12]	53.14 [52.28, 53.99]	28.84 [28.46, 29.22]
MDETR-R101	FineCops val	S/E	65.56 [64.35, 66.73]	61.49 [60.70, 62.26]	66.54 [65.31, 67.79]	86.50 [85.61, 87.43]	54.68 [53.71, 55.64]	29.52 [29.10, 29.92]
MDETR-R101	FineCops val	S/Y	65.56 [64.35, 66.73]	56.10 [55.27, 56.94]	81.17 [80.14, 82.16]	91.61 [90.85, 92.45]	53.10 [52.20, 53.95]	28.65 [28.27, 29.03]
MDETR-R101	FineCops val	Product	65.56 [64.35, 66.73]	60.52 [59.76, 61.26]	69.52 [68.40, 70.65]	87.87 [86.95, 88.80]	54.14 [53.23, 55.06]	29.33 [28.92, 29.73]
MDETR-R101	FineCops val	SIRC-style	65.56 [64.35, 66.73]	52.43 [51.64, 53.25]	74.83 [73.71, 75.95]	92.99 [92.26, 93.68]	57.22 [56.32, 58.12]	30.01 [29.62, 30.39]
MM-GDINO-T	gRef source-disjoint	Native	56.04 [54.42, 57.71]	59.55 [58.54, 60.55]	72.46 [70.87, 73.97]	81.36 [79.59, 83.13]	55.97 [54.66, 57.29]	30.05 [29.50, 30.57]
MM-GDINO-T	gRef source-disjoint	G/E	56.04 [54.42, 57.71]	75.79 [74.89, 76.67]	66.46 [64.78, 68.07]	72.89 [70.16, 75.73]	51.18 [49.90, 52.49]	28.68 [28.13, 29.21]
MM-GDINO-T	gRef source-disjoint	G/Y	56.04 [54.42, 57.71]	70.84 [69.93, 71.72]	72.99 [71.45, 74.51]	70.12 [67.49, 72.70]	50.79 [49.53, 52.07]	28.61 [28.07, 29.15]
MM-GDINO-T	gRef source-disjoint	S/E	56.04 [54.42, 57.71]	76.01 [75.09, 76.89]	66.58 [64.93, 68.18]	70.84 [67.71, 73.08]	51.04 [49.75, 52.34]	28.65 [28.10, 29.19]

Table 9. All score endpoints, $\times 100$: seed means with paired image-cluster intervals. P@1 is identical across confidence routes. Part 2.

Localizer	Population	Score	P@1	Existence	C/W	FPR95	AURC	AUGRC
MM-GDINO-T	gRef source-disjoint	S/Y	56.04 [54.42, 57.71]	70.76 [69.82, 71.67]	72.64 [71.12, 74.13]	71.59 [69.01, 73.75]	50.72 [49.46, 52.00]	28.63 [28.09, 29.17]
MM-GDINO-T	gRef source-disjoint	Product	56.04 [54.42, 57.71]	72.46 [71.56, 73.34]	72.62 [71.02, 74.17]	72.17 [69.11, 75.11]	50.67 [49.41, 51.92]	28.48 [27.94, 29.01]
MM-GDINO-T	gRef source-disjoint	SIRC-style	56.04 [54.42, 57.71]	70.43 [69.50, 71.32]	73.43 [71.82, 74.99]	72.71 [69.87, 75.46]	52.43 [51.19, 53.71]	28.71 [28.18, 29.25]
MDETR-R101	gRef source-disjoint	Native	81.98 [80.32, 83.51]	65.35 [64.27, 66.37]	83.52 [81.94, 85.05]	88.32 [86.68, 90.56]	37.09 [36.14, 38.07]	21.13 [20.68, 21.59]
MDETR-R101	gRef source-disjoint	G/E	81.98 [80.32, 83.51]	72.35 [71.28, 73.36]	78.55 [76.54, 80.48]	82.13 [79.26, 85.97]	33.89 [32.97, 34.83]	20.11 [19.65, 20.57]
MDETR-R101	gRef source-disjoint	G/Y	81.98 [80.32, 83.51]	70.50 [69.36, 71.58]	84.75 [83.23, 86.22]	87.07 [84.44, 89.46]	32.68 [31.85, 33.55]	19.92 [19.48, 20.38]
MDETR-R101	gRef source-disjoint	S/E	81.98 [80.32, 83.51]	70.88 [69.76, 71.92]	78.75 [76.68, 80.73]	86.04 [83.22, 88.47]	34.24 [33.31, 35.19]	20.34 [19.89, 20.81]
MDETR-R101	gRef source-disjoint	S/Y	81.98 [80.32, 83.51]	68.60 [67.44, 69.71]	83.74 [82.08, 85.36]	88.76 [86.54, 90.80]	33.80 [32.90, 34.71]	20.35 [19.90, 20.82]
MDETR-R101	gRef source-disjoint	Product	81.98 [80.32, 83.51]	72.74 [71.63, 73.79]	80.69 [78.81, 82.50]	83.14 [79.66, 86.81]	32.88 [32.00, 33.79]	19.84 [19.39, 20.31]
MDETR-R101	gRef source-disjoint	SIRC-style	81.98 [80.32, 83.51]	66.01 [64.90, 67.05]	84.18 [82.58, 85.71]	87.25 [85.19, 89.40]	36.65 [35.71, 37.61]	20.95 [20.50, 21.40]
MM-GDINO-T	gRef Full	Native	55.95 [54.52, 57.47]	59.41 [58.47, 60.34]	71.79 [70.37, 73.11]	81.19 [79.65, 82.61]	56.38 [55.18, 57.55]	30.24 [29.76, 30.72]
MM-GDINO-T	gRef Full	G/E	55.95 [54.52, 57.47]	75.58 [74.75, 76.41]	66.52 [65.03, 67.97]	73.13 [70.61, 75.82]	51.35 [50.20, 52.52]	28.79 [28.31, 29.28]
MM-GDINO-T	gRef Full	G/Y	55.95 [54.52, 57.47]	70.75 [69.91, 71.56]	72.71 [71.32, 74.00]	70.18 [67.63, 72.50]	51.06 [49.90, 52.22]	28.75 [28.26, 29.23]
MM-GDINO-T	gRef Full	S/E	55.95 [54.52, 57.47]	75.75 [74.91, 76.61]	66.66 [65.18, 68.11]	71.11 [68.16, 73.32]	51.22 [50.08, 52.40]	28.76 [28.28, 29.25]
MM-GDINO-T	gRef Full	S/Y	55.95 [54.52, 57.47]	70.70 [69.82, 71.53]	72.43 [71.07, 73.72]	71.68 [69.43, 73.67]	50.96 [49.80, 52.12]	28.76 [28.28, 29.25]
MM-GDINO-T	gRef Full	Product	55.95 [54.52, 57.47]	72.25 [71.40, 73.08]	72.51 [71.09, 73.84]	72.25 [69.35, 74.99]	50.94 [49.79, 52.08]	28.62 [28.14, 29.10]
MM-GDINO-T	gRef Full	SIRC-style	55.95 [54.52, 57.47]	70.18 [69.35, 71.02]	73.17 [71.71, 74.51]	72.82 [70.11, 75.59]	52.75 [51.61, 53.86]	28.86 [28.38, 29.34]
MDETR-R101	gRef Full	Native	81.84 [80.41, 83.31]	65.22 [64.30, 66.18]	83.41 [81.95, 84.82]	88.30 [86.69, 90.24]	37.43 [36.59, 38.31]	21.27 [20.86, 21.69]

Table 10. All score endpoints, $\times 100$: seed means with paired image-cluster intervals. P@1 is identical across confidence routes. Part 3.

Localizer	Population	Score	P@1	Existence	C/W	FPR95	AURC	AUGRC
MDETR-R101	gRef Full	G/E	81.84 [80.41, 83.31]	72.28 [71.34, 73.26]	78.32 [76.49, 80.13]	81.64 [79.12, 84.62]	34.22 [33.39, 35.08]	20.25 [19.84, 20.67]
MDETR-R101	gRef Full	G/Y	81.84 [80.41, 83.31]	70.49 [69.50, 71.51]	84.07 [82.47, 85.58]	86.17 [83.95, 88.46]	33.10 [32.33, 33.91]	20.09 [19.69, 20.52]
MDETR-R101	gRef Full	S/E	81.84 [80.41, 83.31]	70.68 [69.70, 71.70]	78.44 [76.49, 80.31]	85.80 [83.30, 88.05]	34.70 [33.83, 35.59]	20.52 [20.10, 20.94]
MDETR-R101	gRef Full	S/Y	81.84 [80.41, 83.31]	68.57 [67.55, 69.63]	83.08 [81.42, 84.66]	88.58 [86.55, 90.38]	34.20 [33.36, 35.06]	20.53 [20.12, 20.95]
MDETR-R101	gRef Full	Product	81.84 [80.41, 83.31]	72.65 [71.69, 73.66]	80.47 [78.76, 82.16]	82.43 [79.37, 85.61]	33.25 [32.44, 34.10]	19.99 [19.58, 20.42]
MDETR-R101	gRef Full	SIRC-style	81.84 [80.41, 83.31]	65.90 [64.95, 66.88]	84.05 [82.58, 85.48]	87.01 [85.04, 88.93]	37.00 [36.15, 37.87]	21.09 [20.68, 21.50]

Table 11. FineCops val error composition at requested coverage 50/90%. Achieved coverage is the equal-seed mean fraction of all requests retained, in percent; the boundary tie group is accepted in full. W, N and mixed are conditional error probabilities among accepted requests, $\times 100$, with paired image-cluster 95% intervals. They are not counts or failure rates divided by all requests. W+N=mixed before display rounding. No deployment threshold is fitted; all other coverages/surfaces remain in the full analysis JSON.

Localizer	Head	Requested (%)	Achieved (%)	W|accept	N|accept	Failure|accept
MM-GDINO-T	G/E	50	50.003	8.44 [7.67, 9.25]	41.46 [40.45, 42.40]	49.90 [48.97, 50.76]
MM-GDINO-T	G/E	90	90.003	12.00 [11.25, 12.77]	45.26 [44.33, 46.13]	57.25 [56.50, 57.95]
MM-GDINO-T	G/Y	50	50.003	3.55 [3.13, 4.03]	46.81 [45.95, 47.65]	50.36 [49.53, 51.23]
MM-GDINO-T	G/Y	90	90.003	11.04 [10.34, 11.80]	46.35 [45.45, 47.20]	57.39 [56.68, 58.08]
MM-GDINO-T	S/E	50	50.003	8.41 [7.67, 9.19]	41.45 [40.47, 42.39]	49.87 [48.92, 50.73]
MM-GDINO-T	S/E	90	90.003	11.98 [11.23, 12.76]	45.25 [44.34, 46.13]	57.23 [56.50, 57.93]
MM-GDINO-T	S/Y	50	50.003	3.83 [3.38, 4.36]	46.45 [45.56, 47.28]	50.28 [49.43, 51.14]
MM-GDINO-T	S/Y	90	90.003	10.85 [10.13, 11.58]	46.69 [45.80, 47.53]	57.54 [56.81, 58.21]
MDETR-R101	G/E	50	50.003	15.38 [14.52, 16.25]	41.06 [40.09, 42.02]	56.44 [55.49, 57.40]
MDETR-R101	G/E	90	90.003	17.74 [16.97, 18.55]	46.35 [45.46, 47.20]	64.09 [63.35, 64.81]
MDETR-R101	G/Y	50	50.003	9.04 [8.38, 9.78]	44.52 [43.55, 45.43]	53.56 [52.67, 54.44]
MDETR-R101	G/Y	90	90.003	16.29 [15.54, 17.07]	47.34 [46.49, 48.19]	63.63 [62.89, 64.37]
MDETR-R101	S/E	50	50.003	14.76 [13.86, 15.66]	40.98 [39.94, 41.99]	55.74 [54.74, 56.69]
MDETR-R101	S/E	90	90.003	17.52 [16.74, 18.32]	46.45 [45.58, 47.31]	63.97 [63.22, 64.69]
MDETR-R101	S/Y	50	50.003	8.02 [7.34, 8.72]	44.91 [43.96, 45.88]	52.94 [52.02, 53.86]
MDETR-R101	S/Y	90	90.003	15.78 [15.02, 16.56]	47.68 [46.81, 48.50]	63.46 [62.71, 64.17]

Table 12. Frozen weights, alternate inference readout: mixed AUGRC $\times 100$. Matched and alternate values are seed mean (sample SD); differences use paired 95% intervals. Every row keeps Native boxes fixed. These off-diagonal readings diagnose the training/deployment readout contract; they are not substituted for the matched four-cell matrix. Part 1.

Localizer	Population	Trained head	Alternate read	Matched (SD)	Alternate (SD)	Alt.−matched [CI]
MM-GDINO-T	FineCops val	G/Y	S	26.38 (0.04)	26.44 (0.03)	+0.058 [+0.019, +0.097]
MM-GDINO-T	FineCops val	G/E	S	26.23 (0.07)	26.25 (0.05)	+0.016 [-0.030, +0.061]
MM-GDINO-T	FineCops val	S/Y	G	26.36 (0.04)	28.38 (0.17)	+2.029 [+1.876, +2.179]
MM-GDINO-T	FineCops val	S/E	G	26.20 (0.07)	27.70 (0.49)	+1.506 [+1.365, +1.645]
MDETR-R101	FineCops val	G/Y	S	28.84 (0.13)	30.45 (2.83)	+1.608 [+1.513, +1.698]
MDETR-R101	FineCops val	G/E	S	29.71 (0.27)	32.51 (2.87)	+2.799 [+2.642, +2.952]
MDETR-R101	FineCops val	S/Y	G	28.65 (0.02)	29.09 (0.08)	+0.435 [+0.371, +0.495]
MDETR-R101	FineCops val	S/E	G	29.52 (0.28)	29.97 (0.24)	+0.449 [+0.361, +0.532]
MM-GDINO-T	gRef source-disjoint	G/Y	S	28.61 (0.04)	28.62 (0.04)	+0.012 [-0.010, +0.036]
MM-GDINO-T	gRef source-disjoint	G/E	S	28.68 (0.08)	28.57 (0.08)	-0.105 [-0.139, -0.069]
MM-GDINO-T	gRef source-disjoint	S/Y	G	28.63 (0.05)	30.50 (0.23)	+1.866 [+1.672, +2.059]
MM-GDINO-T	gRef source-disjoint	S/E	G	28.65 (0.11)	29.91 (0.30)	+1.260 [+1.117, +1.400]
MDETR-R101	gRef source-disjoint	G/Y	S	19.92 (0.38)	22.77 (3.43)	+2.853 [+2.740, +2.963]
MDETR-R101	gRef source-disjoint	G/E	S	20.11 (0.54)	24.47 (3.53)	+4.362 [+4.173, +4.542]
MDETR-R101	gRef source-disjoint	S/Y	G	20.35 (0.06)	20.25 (0.11)	-0.109 [-0.167, -0.049]
MDETR-R101	gRef source-disjoint	S/E	G	20.34 (0.15)	20.40 (0.01)	+0.057 [-0.010, +0.128]
MM-GDINO-T	gRef Full	G/Y	S	28.75 (0.05)	28.77 (0.05)	+0.017 [-0.006, +0.040]
MM-GDINO-T	gRef Full	G/E	S	28.79 (0.08)	28.70 (0.09)	-0.095 [-0.126, -0.063]

Table 13. Frozen weights, alternate inference readout: mixed AUGRC $\times 100$. Matched and alternate values are seed mean (sample SD); differences use paired 95% intervals. Every row keeps Native boxes fixed. These off-diagonal readings diagnose the training/deployment readout contract; they are not substituted for the matched four-cell matrix. Part 2.

Localizer	Population	Trained head	Alternate read	Matched (SD)	Alternate (SD)	Alt.−matched [CI]
MM-GDINO-T	gRef Full	S/Y	G	28.76 (0.05)	30.61 (0.22)	+1.851 [+1.680, +2.024]
MM-GDINO-T	gRef Full	S/E	G	28.76 (0.11)	30.02 (0.29)	+1.254 [+1.125, +1.377]
MDETR-R101	gRef Full	G/Y	S	20.09 (0.39)	22.92 (3.37)	+2.834 [+2.735, +2.935]
MDETR-R101	gRef Full	G/E	S	20.25 (0.53)	24.62 (3.52)	+4.369 [+4.202, +4.533]
MDETR-R101	gRef Full	S/Y	G	20.53 (0.06)	20.42 (0.11)	-0.105 [-0.159, -0.053]
MDETR-R101	gRef Full	S/E	G	20.52 (0.14)	20.55 (0.03)	+0.034 [-0.027, +0.097]

Table 14. Same-image state comparisons: exact participating images, requests and within-image pairs. The conditional and comparable-unconditional estimates use these same participating requests; bootstrap image multiplicity applies once to each within-copy pair.

Localizer	Population	State pair	Images	Requests	Within-image pairs
MM-GDINO-T	FineCops val	CW	835	3943	5016
MM-GDINO-T	FineCops val	CN	2344	14749	32776
MM-GDINO-T	FineCops val	WN	624	4040	6110
MDETR-R101	FineCops val	CW	1177	5340	6921
MDETR-R101	FineCops val	CN	2194	13350	26598
MDETR-R101	FineCops val	WN	1088	7322	12288
MM-GDINO-T	gRef source-disjoint	CW	1044	8322	15552
MM-GDINO-T	gRef source-disjoint	CN	1215	12871	33120
MM-GDINO-T	gRef source-disjoint	WN	1103	10979	25726
MDETR-R101	gRef source-disjoint	CW	584	5198	8744
MDETR-R101	gRef source-disjoint	CN	1266	15723	48801
MDETR-R101	gRef source-disjoint	WN	593	5326	10045
MM-GDINO-T	gRef Full	CW	1232	9808	18448
MM-GDINO-T	gRef Full	CN	1430	15171	39160
MM-GDINO-T	gRef Full	WN	1299	12973	30428
MDETR-R101	gRef Full	CW	692	6166	10397
MDETR-R101	gRef Full	CN	1488	18517	57647
MDETR-R101	gRef Full	WN	702	6351	11941

Table 15. FineCops original-positive-difficulty counts. High/low states are the first/second state in the pair label. Zero low-state counts at L2/L3 expose the absence of same-level no-target comparisons; these cells are not assigned an AUROC. Images overlap across levels/state comparisons.

Localizer	State pair	Level	High-state requests	Low-state requests	Images
MM-GDINO-T	CW	L1	5506	591	2710
MM-GDINO-T	CW	L2	1566	1318	1332
MM-GDINO-T	CW	L3	220	225	269
MM-GDINO-T	CN	L1	5506	9029	2686
MM-GDINO-T	CN	L2	1566	0	966
MM-GDINO-T	CN	L3	220	0	168
MM-GDINO-T	WN	L1	591	9029	2458
MM-GDINO-T	WN	L2	1318	0	805
MM-GDINO-T	WN	L3	225	0	167
MDETR-R101	CW	L1	4581	1516	2710
MDETR-R101	CW	L2	1393	1491	1332
MDETR-R101	CW	L3	206	239	269
MDETR-R101	CN	L1	4581	9029	2673
MDETR-R101	CN	L2	1393	0	909
MDETR-R101	CN	L3	206	0	157
MDETR-R101	WN	L1	1516	9029	2476
MDETR-R101	WN	L2	1491	0	889
MDETR-R101	WN	L3	239	0	172

Table 16. Conditional effects and their uncertainty, AUROC $\times 100$. Each cell is mean [paired 95% CI], not a raw score AUROC. $Y : S - G$ is the emission readout effect; $G : Y - E$ is the global target effect. Image conditioning matches participating requests exactly. Level conditioning changes the available within-level pair composition (including the zero cells in the count table), not a difficulty-matched target distribution. An undefined interval is not zero effect or proof of attribution. Part 1.

Localizer	Population	Effect	Condition	Comparable uncond.	Conditional	Uncond.−cond.
MM-GDINO-T	FineCops val	$Y : S - G$	Image CW	-0.405 [-0.822, +0.012]	+0.159 [-0.661, +0.916]	-0.565 [-1.219, +0.148]
MM-GDINO-T	FineCops val	$Y : S - G$	Level CW	-0.600 [-0.829, -0.372]	-0.591 [-0.889, -0.299]	-0.009 [-0.156, +0.138]
MM-GDINO-T	FineCops val	$G : Y - E$	Image CW	+8.854 [+7.523, +10.221]	+7.642 [+5.850, +9.421]	+1.211 [-0.222, +2.634]
MM-GDINO-T	FineCops val	$G : Y - E$	Level CW	+11.654 [+10.721, +12.641]	+6.940 [+5.993, +7.944]	+4.713 [+4.075, +5.363]
MM-GDINO-T	FineCops val	$Y : S - G$	Image CN	+0.241 [+0.129, +0.349]	+0.289 [+0.066, +0.512]	-0.048 [-0.231, +0.139]
MM-GDINO-T	FineCops val	$Y : S - G$	Level CN	+0.261 [+0.154, +0.366]	+0.216 [+0.112, +0.317]	+0.045 [-0.020, +0.109]
MM-GDINO-T	FineCops val	$G : Y - E$	Image CN	-1.700 [-2.002, -1.400]	-1.163 [-1.557, -0.760]	-0.537 [-0.877, -0.202]
MM-GDINO-T	FineCops val	$G : Y - E$	Level CN	-3.523 [-3.884, -3.166]	+0.224 [+0.024, +0.431]	-3.747 [-4.076, -3.434]
MM-GDINO-T	FineCops val	$Y : S - G$	Image WN	+0.397 [-0.016, +0.792]	+0.660 [-0.026, +1.326]	-0.263 [-0.854, +0.331]
MM-GDINO-T	FineCops val	$Y : S - G$	Level WN	+0.619 [+0.366, +0.863]	+0.220 [-0.117, +0.540]	+0.399 [+0.085, +0.719]
MM-GDINO-T	FineCops val	$G : Y - E$	Image WN	-12.753 [-14.160, -11.363]	-10.644 [-12.590, -8.700]	-2.109 [-3.612, -0.580]
MM-GDINO-T	FineCops val	$G : Y - E$	Level WN	-17.795 [-18.821, -16.781]	-8.720 [-9.994, -7.505]	-9.075 [-10.290, -7.813]
MDETR-R101	FineCops val	$Y : S - G$	Image CW	+2.986 [+2.370, +3.600]	+2.712 [+1.668, +3.786]	+0.275 [-0.529, +1.106]
MDETR-R101	FineCops val	$Y : S - G$	Level CW	+2.373 [+1.929, +2.839]	+2.586 [+2.093, +3.107]	-0.213 [-0.434, +0.000]
MDETR-R101	FineCops val	$G : Y - E$	Image CW	+12.180 [+11.106, +13.269]	+11.752 [+10.280, +13.309]	+0.429 [-0.665, +1.493]
MDETR-R101	FineCops val	$G : Y - E$	Level CW	+13.889 [+13.044, +14.743]	+13.660 [+12.738, +14.596]	+0.229 [-0.192, +0.661]
MDETR-R101	FineCops val	$Y : S - G$	Image CN	+0.727 [+0.392, +1.066]	+0.924 [+0.410, +1.456]	-0.197 [-0.587, +0.206]
MDETR-R101	FineCops val	$Y : S - G$	Level CN	+0.301 [-0.016, +0.622]	+0.834 [+0.514, +1.166]	-0.533 [-0.719, -0.349]

Table 17. Conditional effects and their uncertainty, AUROC $\times 100$. Each cell is mean [paired 95% CI], not a raw score AUROC. $Y : S - G$ is the emission readout effect; $G : Y - E$ is the global target effect. Image conditioning matches participating requests exactly. Level conditioning changes the available within-level pair composition (including the zero cells in the count table), not a difficulty-matched target distribution. An undefined interval is not zero effect or proof of attribution. Part 2.

Localizer	Population	Effect	Condition	Comparable uncond.	Conditional	Uncond.−cond.
MDETR-R101	FineCops val	$G : Y - E$	Image CN	+0.450 [-0.121, +1.027]	+1.307 [+0.509, +2.098]	-0.857 [-1.453, -0.265]
MDETR-R101	FineCops val	$G : Y - E$	Level CN	+0.290 [-0.265, +0.857]	+2.195 [+1.647, +2.750]	-1.905 [-2.213, -1.595]
MDETR-R101	FineCops val	$Y : S - G$	Image WN	-3.433 [-4.085, -2.798]	-2.772 [-3.850, -1.762]	-0.661 [-1.466, +0.165]
MDETR-R101	FineCops val	$Y : S - G$	Level WN	-2.696 [-3.193, -2.211]	-3.157 [-3.834, -2.501]	+0.461 [+0.001, +0.954]
MDETR-R101	FineCops val	$G : Y - E$	Image WN	-10.839 [-11.825, -9.875]	-9.608 [-11.090, -8.055]	-1.230 [-2.362, -0.072]
MDETR-R101	FineCops val	$G : Y - E$	Level WN	-14.257 [-15.124, -13.406]	-13.524 [-14.635, -12.448]	-0.733 [-1.527, +0.055]
MM-GDINO-T	gRef source-disjoint	$Y : S - G$	Image CW	-0.378 [-0.672, -0.095]	-0.039 [-0.829, +0.750]	-0.340 [-1.047, +0.387]
MM-GDINO-T	gRef source-disjoint	$G : Y - E$	Image CW	+6.070 [+5.150, +6.975]	+3.920 [+2.668, +5.218]	+2.150 [+1.000, +3.341]
MM-GDINO-T	gRef source-disjoint	$Y : S - G$	Image CN	+0.036 [-0.114, +0.185]	+0.430 [+0.164, +0.701]	-0.394 [-0.607, -0.179]
MM-GDINO-T	gRef source-disjoint	$G : Y - E$	Image CN	-3.316 [-3.764, -2.885]	-4.266 [-4.832, -3.717]	+0.950 [+0.500, +1.404]
MM-GDINO-T	gRef source-disjoint	$Y : S - G$	Image WN	-0.258 [-0.529, +0.003]	-0.277 [-0.767, +0.196]	+0.019 [-0.384, +0.408]
MM-GDINO-T	gRef source-disjoint	$G : Y - E$	Image WN	-6.775 [-7.481, -6.061]	-5.456 [-6.318, -4.548]	-1.319 [-2.107, -0.578]
MDETR-R101	gRef source-disjoint	$Y : S - G$	Image CW	-1.301 [-2.039, -0.532]	-1.616 [-2.740, -0.541]	+0.315 [-0.616, +1.282]
MDETR-R101	gRef source-disjoint	$G : Y - E$	Image CW	+4.478 [+3.273, +5.734]	+7.087 [+5.545, +8.782]	-2.608 [-3.920, -1.291]
MDETR-R101	gRef source-disjoint	$Y : S - G$	Image CN	-1.927 [-2.211, -1.634]	-1.540 [-1.914, -1.175]	-0.386 [-0.664, -0.111]
MDETR-R101	gRef source-disjoint	$G : Y - E$	Image CN	-0.509 [-0.934, -0.098]	+1.155 [+0.680, +1.617]	-1.664 [-2.050, -1.278]
MDETR-R101	gRef source-disjoint	$Y : S - G$	Image WN	-2.145 [-3.106, -1.217]	-2.150 [-3.233, -1.075]	+0.005 [-0.804, +0.855]
MDETR-R101	gRef source-disjoint	$G : Y - E$	Image WN	-5.580 [-7.305, -4.013]	-5.044 [-6.793, -3.358]	-0.536 [-1.760, +0.681]

Table 18. Conditional effects and their uncertainty, AUROC $\times 100$. Each cell is mean [paired 95% CI], not a raw score AUROC. $Y : S-G$ is the emission readout effect; $G : Y-E$ is the global target effect. Image conditioning matches participating requests exactly. Level conditioning changes the available within-level pair composition (including the zero cells in the count table), not a difficulty-matched target distribution. An undefined interval is not zero effect or proof of attribution. Part 3.

Localizer	Population	Effect	Condition	Comparable uncond.	Conditional	Uncond.−cond.
MM-GDINO-T	gRef Full	$Y : S-G$	Image CW	-0.286 [-0.559, -0.014]	-0.069 [-0.783, +0.641]	-0.217 [-0.846, +0.447]
MM-GDINO-T	gRef Full	$G : Y-E$	Image CW	+5.685 [+4.847, +6.520]	+3.905 [+2.740, +5.049]	+1.780 [+0.701, +2.839]
MM-GDINO-T	gRef Full	$Y : S-G$	Image CN	+0.077 [-0.067, +0.211]	+0.375 [+0.116, +0.629]	-0.299 [-0.500, -0.094]
MM-GDINO-T	gRef Full	$G : Y-E$	Image CN	-3.276 [-3.676, -2.861]	-4.199 [-4.727, -3.663]	+0.923 [+0.523, +1.342]
MM-GDINO-T	gRef Full	$Y : S-G$	Image WN	-0.244 [-0.485, +0.000]	-0.348 [-0.792, +0.085]	+0.104 [-0.250, +0.453]
MM-GDINO-T	gRef Full	$G : Y-E$	Image WN	-6.524 [-7.173, -5.869]	-5.531 [-6.309, -4.721]	-0.993 [-1.692, -0.298]
MDETR-R101	gRef Full	$Y : S-G$	Image CW	-1.217 [-1.916, -0.546]	-1.552 [-2.567, -0.611]	+0.335 [-0.450, +1.193]
MDETR-R101	gRef Full	$G : Y-E$	Image CW	+4.117 [+2.833, +5.388]	+6.620 [+5.146, +8.235]	-2.503 [-3.693, -1.282]
MDETR-R101	gRef Full	$Y : S-G$	Image CN	-1.946 [-2.205, -1.669]	-1.689 [-2.013, -1.342]	-0.257 [-0.508, -0.009]
MDETR-R101	gRef Full	$G : Y-E$	Image CN	-0.534 [-0.914, -0.138]	+1.051 [+0.593, +1.497]	-1.585 [-1.933, -1.227]
MDETR-R101	gRef Full	$Y : S-G$	Image WN	-2.245 [-3.087, -1.451]	-2.116 [-3.190, -1.050]	-0.129 [-0.975, +0.733]
MDETR-R101	gRef Full	$G : Y-E$	Image WN	-5.285 [-6.869, -3.778]	-4.958 [-6.420, -3.450]	-0.328 [-1.447, +0.778]

Table 19. True edited-negative/positive-parent pair wins on FineCops, $\times 100$. Raw columns are equal-seed mean pair wins; effects include paired image-cluster 95% intervals. C/W refers to the parent's Native correctness. Pair counts repeat positive parents; C/W image counts may overlap. These are diagnostic pair wins, not official Recall@1.

Localizer	Parent	Pairs (images)	G/E	G/Y	S/Y	$G : Y-E$	$Y : S-G$
MM-GDINO-T	all	9029 (2433)	67.09	66.60	66.42	-0.495 [-0.986, -0.027]	-0.174 [-0.563, +0.196]
MM-GDINO-T	C	8136 (2303)	67.99	67.76	67.52	-0.229 [-0.651, +0.208]	-0.234 [-0.606, +0.134]
MM-GDINO-T	W	893 (348)	58.94	56.03	56.40	-2.912 [-5.905, +0.000]	+0.373 [-1.407, +2.114]
MDETR-R101	all	9029 (2433)	64.52	62.83	62.96	-1.683 [-2.562, -0.823]	+0.126 [-0.557, +0.811]
MDETR-R101	C	6675 (2092)	66.78	67.92	69.20	+1.144 [+0.187, +2.068]	+1.283 [+0.548, +2.026]
MDETR-R101	W	2354 (845)	58.11	48.41	45.26	-9.700 [-11.789, -7.615]	-3.158 [-4.789, -1.644]

Table 20. Dense-logit global-winner geometry, $\times 100$; seed-mean rates/IoU, not causal spatial contributions. For G-trained heads this winner supplies matched confidence. For S-trained heads it is only a counterfactual diagnostic: matched S deployment always reads the Native query and has zero readout-index disagreement. Winner-correct is shown only for Native-wrong positives. Part 1.

Localizer	Population	Head	State	Requests	Query differs	Box IoU	Winner correct
MM-GDINO-T	FineCops val	G/E	C	7292	7.94	93.94	–
MM-GDINO-T	FineCops val	G/E	W	2134	29.94	72.01	14.18
MM-GDINO-T	FineCops val	G/E	N	9029	10.40	92.79	–
MM-GDINO-T	FineCops val	G/Y	C	7292	4.85	96.71	–
MM-GDINO-T	FineCops val	G/Y	W	2134	18.20	83.25	8.83
MM-GDINO-T	FineCops val	G/Y	N	9029	6.65	96.13	–
MM-GDINO-T	FineCops val	S/E	C	7292	48.04	84.23	–
MM-GDINO-T	FineCops val	S/E	W	2134	66.67	55.43	16.59
MM-GDINO-T	FineCops val	S/E	N	9029	66.60	78.92	–
MM-GDINO-T	FineCops val	S/Y	C	7292	59.94	86.40	–
MM-GDINO-T	FineCops val	S/Y	W	2134	84.88	60.79	17.37
MM-GDINO-T	FineCops val	S/Y	N	9029	72.06	82.22	–
MDETR-R101	FineCops val	G/E	C	6180	72.21	42.33	–
MDETR-R101	FineCops val	G/E	W	3246	81.91	31.42	8.93
MDETR-R101	FineCops val	G/E	N	9029	77.17	36.64	–
MDETR-R101	FineCops val	G/Y	C	6180	58.14	63.87	–
MDETR-R101	FineCops val	G/Y	W	3246	62.62	53.96	8.88
MDETR-R101	FineCops val	G/Y	N	9029	58.45	60.37	–
MDETR-R101	FineCops val	S/E	C	6180	41.49	69.26	–
MDETR-R101	FineCops val	S/E	W	3246	65.10	48.61	17.74
MDETR-R101	FineCops val	S/E	N	9029	57.25	56.50	–
MDETR-R101	FineCops val	S/Y	C	6180	15.72	89.32	–
MDETR-R101	FineCops val	S/Y	W	3246	54.64	55.69	19.42
MDETR-R101	FineCops val	S/Y	N	9029	32.62	75.15	–

Table 21. Dense-logit global-winner geometry, $\times 100$; seed-mean rates/IoU, not causal spatial contributions. For G-trained heads this winner supplies matched confidence. For S-trained heads it is only a counterfactual diagnostic: matched S deployment always reads the Native query and has zero readout-index disagreement. Winner-correct is shown only for Native-wrong positives. Part 2.

Localizer	Population	Head	State	Requests	Query differs	Box IoU	Winner correct
MM-GDINO-T	gRef source-disjoint	G/E	C	5519	15.14	86.57	–
MM-GDINO-T	gRef source-disjoint	G/E	W	4329	33.65	69.90	10.98
MM-GDINO-T	gRef source-disjoint	G/E	N	7716	34.37	70.80	–
MM-GDINO-T	gRef source-disjoint	G/Y	C	5519	8.61	92.61	–
MM-GDINO-T	gRef source-disjoint	G/Y	W	4329	22.58	80.74	8.39
MM-GDINO-T	gRef source-disjoint	G/Y	N	7716	23.52	81.38	–
MM-GDINO-T	gRef source-disjoint	S/E	C	5519	55.92	63.45	–
MM-GDINO-T	gRef source-disjoint	S/E	W	4329	77.99	43.94	12.10
MM-GDINO-T	gRef source-disjoint	S/E	N	7716	88.02	40.75	–
MM-GDINO-T	gRef source-disjoint	S/Y	C	5519	70.22	70.72	–
MM-GDINO-T	gRef source-disjoint	S/Y	W	4329	92.22	52.96	16.01
MM-GDINO-T	gRef source-disjoint	S/Y	N	7716	92.12	52.06	–
MDETR-R101	gRef source-disjoint	G/E	C	8073	71.09	46.48	–
MDETR-R101	gRef source-disjoint	G/E	W	1775	80.11	34.12	10.35
MDETR-R101	gRef source-disjoint	G/E	N	7716	77.06	38.10	–
MDETR-R101	gRef source-disjoint	G/Y	C	8073	66.26	63.31	–
MDETR-R101	gRef source-disjoint	G/Y	W	1775	62.25	57.50	8.30
MDETR-R101	gRef source-disjoint	G/Y	N	7716	63.57	60.09	–
MDETR-R101	gRef source-disjoint	S/E	C	8073	32.08	78.03	–
MDETR-R101	gRef source-disjoint	S/E	W	1775	64.09	50.90	17.71
MDETR-R101	gRef source-disjoint	S/E	N	7716	56.70	56.49	–
MDETR-R101	gRef source-disjoint	S/Y	C	8073	19.08	87.00	–
MDETR-R101	gRef source-disjoint	S/Y	W	1775	51.55	59.35	11.44
MDETR-R101	gRef source-disjoint	S/Y	N	7716	37.61	71.66	–

Table 22. Dense-logit global-winner geometry, ×100; seed-mean rates/IoU, not causal spatial contributions. For G-trained heads this winner supplies matched confidence. For S-trained heads it is only a counterfactual diagnostic: matched S deployment always reads the Native query and has zero readout-index disagreement. Winner-correct is shown only for Native-wrong positives. Part 3.

Localizer	Population	Head	State	Requests	Query differs	Box IoU	Winner correct
MM-GDINO-T	gRef Full	G/E	C	6469	15.22	86.46	–
MM-GDINO-T	gRef Full	G/E	W	5094	33.08	70.46	10.89
MM-GDINO-T	gRef Full	G/E	N	9121	34.13	71.00	–
MM-GDINO-T	gRef Full	G/Y	C	6469	8.69	92.49	–
MM-GDINO-T	gRef Full	G/Y	W	5094	22.01	81.18	8.24
MM-GDINO-T	gRef Full	G/Y	N	9121	23.31	81.47	–
MM-GDINO-T	gRef Full	S/E	C	6469	55.72	63.43	–
MM-GDINO-T	gRef Full	S/E	W	5094	77.76	44.13	12.05
MM-GDINO-T	gRef Full	S/E	N	9121	87.64	40.92	–
MM-GDINO-T	gRef Full	S/Y	C	6469	70.06	70.48	–
MM-GDINO-T	gRef Full	S/Y	W	5094	92.08	52.91	15.61
MM-GDINO-T	gRef Full	S/Y	N	9121	92.02	52.16	–
MDETR-R101	gRef Full	G/E	C	9463	71.11	46.38	–
MDETR-R101	gRef Full	G/E	W	2100	80.03	34.32	10.29
MDETR-R101	gRef Full	G/E	N	9121	76.98	38.19	–
MDETR-R101	gRef Full	G/Y	C	9463	66.28	63.29	–
MDETR-R101	gRef Full	G/Y	W	2100	62.37	57.42	8.30
MDETR-R101	gRef Full	G/Y	N	9121	63.59	59.98	–
MDETR-R101	gRef Full	S/E	C	9463	32.31	77.86	–
MDETR-R101	gRef Full	S/E	W	2100	62.90	51.92	17.37
MDETR-R101	gRef Full	S/E	N	9121	56.45	56.56	–
MDETR-R101	gRef Full	S/Y	C	9463	19.05	87.02	–
MDETR-R101	gRef Full	S/Y	W	2100	50.24	60.50	11.48
MDETR-R101	gRef Full	S/Y	N	9121	37.60	71.56	–

Table 23. AUGRC crossover prior (percent). Interior-only intervals are conditional summaries; non-interior replicates remain counted out of 5,000. Part 1.

Localizer	Population	Contrast	Prior	Root status	All-draw CI	Interior-only CI	Interior draws
MM-GDINO-T	FineCops val	global_emit_minus_exists	42.82	interior	[39.32, 46.40]	[39.32, 46.40]	5000
MM-GDINO-T	FineCops val	selected_emit_minus_exists	41.95	interior	[38.57, 45.56]	[38.57, 45.56]	5000
MM-GDINO-T	FineCops val	D_emit	34.23	interior	[21.42, 51.00]	[21.42, 51.00]	5000
MDETR-R101	FineCops val	global_emit_minus_exists	–	no interior root	–	[91.11, 99.88]	759
MDETR-R101	FineCops val	selected_emit_minus_exists	–	no interior root	–	[88.00, 99.80]	2372
MDETR-R101	FineCops val	D_emit	–	no interior root	–	[76.87, 99.86]	157
MM-GDINO-T	gRef source-disjoint	global_emit_minus_exists	47.42	interior	[41.70, 53.26]	[41.70, 53.26]	5000
MM-GDINO-T	gRef source-disjoint	selected_emit_minus_exists	44.87	interior	[39.16, 50.57]	[39.16, 50.57]	5000
MM-GDINO-T	gRef source-disjoint	D_emit	80.20	interior	–	[22.13, 97.92]	3419
MDETR-R101	gRef source-disjoint	global_emit_minus_exists	69.42	interior	–	[53.22, 91.37]	4955
MDETR-R101	gRef source-disjoint	selected_emit_minus_exists	42.63	interior	[31.18, 58.77]	[31.18, 58.77]	5000
MDETR-R101	gRef source-disjoint	D_emit	–	no interior root	–	[1.28, 1.28]	1
MM-GDINO-T	gRef Full	global_emit_minus_exists	46.28	interior	[40.79, 51.82]	[40.79, 51.82]	5000
MM-GDINO-T	gRef Full	selected_emit_minus_exists	44.30	interior	[38.88, 49.77]	[38.88, 49.77]	5000
MM-GDINO-T	gRef Full	D_emit	61.97	interior	–	[13.87, 96.12]	4282
MDETR-R101	gRef Full	global_emit_minus_exists	66.71	interior	–	[51.55, 88.32]	4973
MDETR-R101	gRef Full	selected_emit_minus_exists	42.99	interior	[31.46, 58.63]	[31.46, 58.63]	5000
MDETR-R101	gRef Full	D_emit	–	no interior root	–	–	0

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